

The Influence of Microgrid Systems on Rural and Urban Electrification for Community Economic Development



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Introduction

Electricity is an immensely powerful force of nature that has been known to human beings since our ancestors first observed them in the form of thunder and lightning. However, it is only in the past century or so that its true utility has been realized, and today it is ubiquitous enough in developed societies that its existence is only really noticed during the occasional blackout.

Although it is essentially taken for granted in the developed, urban societies where the infrastructure and governance required for its operation exists, there are still many areas where reliable electricity is still a faraway dream. There are a large number of variables that influence this availability, but perhaps the most important of these are the economic conditions of the locations attempting to be electrified.

Electrical infrastructure is incredibly expensive to install, both in the form of large scale power generation plants as well as the power lines that must be built to bring this electricity from the generation location to the individual consumers. However, through the application of ever evolving electricity generation and transportation technology, these previously prohibitively expensive technologies are becoming more and more available in places where previously it was not economically viable to do so. One of these emerging technologies are microgrids, an exciting and futuristic way in which energy production can be decentralized and generated locally.

Microgrids have great potential in raising the quality of life of a huge number of individuals, but in the process will bring with it a drastically different economic landscape surrounding energy generation and distribution.

Since the concept of islanded (disconnected from the main power grid) microgrids and microgrids capable of integrating with larger main grids and selling energy back to the grid are finally becoming feasible for large-scale rollout of this technology, there is limited but enthusiastic research being conducted in many different places around the world. In this review, I will analyze a selection of this research in hopes of presenting an understanding of what sort of influence this new alteration to the energy generation economic landscape is predicted to have, and how that influence will extend to community economic development in both developed, developing, and pre-development societies.

Such research can be seen on full display in "Sustainable Microgrids: Economic, Environmental, and Social Costs and Benefits of Microgrid Deployment," (Parag, 2019) which addresses the comparative costs and benefits of microgrid development versus traditional generation connected to bulk transmission and distribution grids. The study focuses on various aspects: environmental (avoiding environmental damage costs), economic (mainly employment multiplier effects), deferral or avoidance of transmission and distribution investment costs, and access to highly reliable and resilient electricity supply.

The context of the study is the Israeli electricity market, characterized by a highly centralized, vertically integrated electricity company (Israel Electric Corporation IEC) and some legacy distribution companies. This context is relevant because of the significant decline in Israel's transmission and distribution infrastructure investment over recent years, positioning the Israeli market as a useful study for analyzing microgrids as an alternative to large-scale infrastructure investments.

The analysis concludes that, under reasonable assumptions reflecting the current state of microgrid technologies, microgrids could be a viable and cost-effective alternative to additional investments in central-station generation and associated transmission and distribution infrastructure. Many other countries share some of the same concerns that Israel does when it comes to meeting their peoples energy demand, and the country can serve as a good case study for the future to come.

Part 1: The Basic Economics of Energy

Economic principles, while for the most part operating on a set of fairly universal tenants, is nonetheless a protean field of study that sees rapid changes as our technology, society, and environment evolve around us. Our world today is almost unrecognizable from the world of only a hundred years ago, which in turn was just as unrecognizable as the world a hundred years prior to that. To understand the degree of economic change that microgrid technology may catalyze, a rudimentary understanding of our current energy markets is necessary. In our modern world, energy is synonymous with electricity. This electricity can be generated from a large variety of sources, including but not limited to biofuels like wood and plants, fossil fuels like coal, petroleum, and natural gas, nuclear energy, and increasingly, renewable generation methods like wind and solar farms.

This electricity has historically been produced in large centralized power plants, with its fuel being sourced from other outside locations. As our ability to conduct global trade and shipping has increased, our fuel sources come from more and more faraway places in the pursuit of

cheaper energy, leading to a centralized system that most individuals are both entirely reliant on as well as incapable of influencing to any substantial degree. The main electrical grid is the foundation of this centralized system, a massive network of transmission lines, transformers, electricity meters, power plants, and other ancillary components that allow for a consumer to purchase energy in the form of electricity and utilize it to improve their quality of life. This demand side of this energy market is most commonly done by large, governmentally regulated entities, acting as what are called natural monopolies to provide energy to their consumers. They factor in generation costs, fuel transportation costs, upkeep costs, and a myriad of other variables to deliver electricity at a \$/kWh (Kilowatt-Hour) price, which is subject to analysis and scrutiny by regulatory bodies to ensure that this basic utility is being distributed in a non-exploitative manner.

This rate for the most part, is not consistent throughout the year, or even throughout the day. It fluctuates greatly depending on the time of day, the weather, the economic conditions of the countries providing the fuel, and a large number of other factors that necessitates a robust and adaptable market to maximize efficiency for both consumers and suppliers. Here, we reach the first example of how microgrids are already altering the status quo of our energy markets. Since microgrids are capable of local energy generation, they are unshackled from the aforementioned market conditions in a variety of ways. Depending on what types of batteries are incorporated into the system, they are able to generate energy during times when electricity is not needed, and then utilize it during peak hours (For example, after the sun sets, or on a particularly warm day to operate the air conditioning) to save on the increased cost of electricity during these times of high demand. They can sell energy back to the grid if they find their usage is low, energy that

will then make its way to other consumers connected to the main grid thus lowering the amount of energy needing to be generated from the central plant.

When it comes to research into this topic, attention is high. In the paper “Energy management considering multiple power markets and microgrid storage” (Yan, 2023), the researchers made a concentrated effort to understand how different pricing tiers and structures worked to influence the impact that microgrids have on the local energy market. The paper analyzes four distinct tariffs: Dynamic Daily Electricity Price, which varies daily; Capacity-Based Charging, a two-part pricing scheme based on maximum consumer power or local transformer capacity; Macro-Renewable Generation Promoting Ancillary Services, aimed at power peak shaving to support renewable generation; and Distributed Solar Generation Feedback Tariff, with sub-tariffs for clean power generation. The study models the response of electric and cold storage in a microgrid to these tariffs using a non-linear mixed-integer optimization problem. Numerical analyses reveal complex trade-offs in optimizing the total cost of microgrid operation within a diverse market environment, providing valuable insights into the intricacies of microgrid cost management. It makes it abundantly clear that to maximize economic potential, and therefore feasibility, there is no such thing as a catch-all system. There will need to be substantial understanding of the local energy market and its associated market structure to determine the final details of any potential microgrid project.

A similar research paper "The Effects of Capital and Energy Subsidies on the Optimal Design of Microgrid Systems" (Díaz, 2020) investigated the influence of subsidy policies on the design of microgrids in islanded mode. Utilizing a computational model, it assesses how capital and fuel

subsidies impact the levelized cost of electricity (LCOE) in different systems. Sensitivity analyses, considering technical parameters and local meteorological conditions show that subsidies substantially alter the final design of microgrids, especially in rural electrification. Such subsidies will play an essential role in determining the way in which this technology is adopted. Considering the role that governing entities play in the creation of these policies, it is paramount that the economic developers working for them have access to market and economics related data. It is another stark example of the importance of good data and making sure to compile it properly, a concept we have touched upon many times over the course of this semester.

Part 2: Microgrids and Community Development

While the topic of microgrid economics was briefly touched on in the previous section, there still remain a large number of facets to the discussion that require analysis. Where does the initial cost of installation come from? Will municipal, state, or federal governments help to pay this cost? Who will be in charge of the upkeep and pay the wages of the specialized engineers and technicians that will be tasked with this maintenance? While too broad to cover sufficiently in this analysis, one important and increasingly relevant question that warrants exploration is the role of governmental policy when it comes to microgrids. In recent years, countries across the globe are increasingly placing focus on the importance of energy independence and the adoption of green, renewable energy over fossil fuels. Massive amounts of funding and subsidization has already seen allocation for every type of renewable energy project imaginable, and microgrids are starting to catch the attention of the policy makers responsible for distributing this funding.

It is here that we run into our first hiccup when discussing this technology. The truth of the matter is that microgrids, especially robust ones designed for use in developed countries with high energy demand, are incredibly expensive. The Blue Lake Rancheria microgrid cost an estimated 15 million dollars for its deployment, and while it does save anywhere from 250 to 400 thousand USD per year on energy costs (Carter, 2019), the predicted lifespan of the project coupled with ongoing maintenance and upkeep costs make it difficult to make a clear argument for the economic benefits of it. We must then shift our focus away from the quantifiable aspects of potential savings, and instead look at those which can only be qualified. As we have learned over the course of the semester, there are many different avenues in which economic development can take place, not all of them able to be represented by simple values. The cohesion of a community, the kind of leadership available to it, and the willingness of its citizens to embrace change are just a few examples of these, and it is here where we must place our attention in regards to the potential benefits of microgrids.

Community economic development (CED) is a difficult and nebulous goal that is nonetheless at the forefront of almost any governmental institution. As described by Flora and Flora, there are 7 key “capitals” that help influence economic development, each corresponding to a vital indicator of a communities overall wellbeing. While arguing in favor of the aforementioned indicators that have less tangible values attached to them is never as straightforward as pointing to a larger or smaller dollar amount relative to the status quo, it is indeed possible.

The first indicator that one should consider is energy sovereignty. In light of recent events (climate crisis, renewable energy boom, global conflicts), it has been made clear that energy as

utilized in today's society is not as abundantly available as we had imagined at the turn of the century. The feasibility of alternative energy distribution systems are being considered increasingly heavily, which has fueled the publication of much research in this field. In the paper "The impact of policy on microgrid economics: A review" (Milis, 2018), the authors focus their analysis on the impact of carbon taxation, economic incentives and mandatory emissions reduction or mandatory minimum percentage participation of renewables in local generation in regards to microgrids. They find that while analysis on microgrid feasibility is indeed taking place, there exist glaring holes in their arguments. Analysis tends to minimize the impact of O and M (Operation and Maintenance) of potential systems, do not clearly define the scale of the microgrid models being created, and that purely renewable based microgrids are economically infeasible and therefore unlikely to lead to greater penetration of renewable based energy, an argument frequently cited when arguing in favor of this technology. It goes to show that there are still many ways in which improvements and refinements can be made to both the technical systems as well as their integration into larger ecosystems, and is in my personal opinion a sign for great optimism.

Another difficult to measure indicator of community benefit can be seen in the general health of its population. It is difficult to quantify healthcare costs and the knock-on effects of reliable healthcare, but it is indisputable that a physically healthier community is preferable to development than an unhealthy one. In the "Demonstrating a Secure, Reliable Low-Carbon Community Microgrid at the Blue Lake Rancheria" report (Carter, 2019) it is detailed how the microgrids can promote general public wellbeing and health at the Blue Lake Rancheria (BLR). The microgrid enhances resilience in sectors such as water, food, transportation, and

communication/IT, which in times of crisis are often the cause of great suffering. The microgrid facilitates the operation of a certified American Red Cross emergency shelter and supports the Tribe's public safety divisions like Tribal Police and the Office of Emergency Services, as well as enabling the Tribe to manage power needs efficiently, supplying critical 'life, health, and safety level power' during emergencies. Case studies like the above can serve as valuable proof of concepts to other communities with similar characteristics across not just the US, but potentially the world.

In part 1, we discussed a few examples of how microgrid integration can be influenced by the energy market and related forces. While this does provide an avenue for policy makers to guide the deployment of microgrid technologies, there is also a more direct method of utilizing them to their full effectiveness. The idea of the paper titled "Welfare Maximization-Based Distributed Demand Response for Islanded Multi-Microgrid Networks Using Diffusion Strategy" (Ali, 2019) proposes a distributed demand response program for islanded multi-microgrid networks, not relying on market price signals. This program focuses on welfare maximization, where optimal power allocation to microgrids occurs in two steps. First, the total surplus and shortage power of the network are determined using local data from each microgrid. Then, surplus power is allocated to microgrids with power shortages based on their welfare functions. The method utilizes a diffusion strategy in both steps, showing effectiveness in simulation results compared to the consensus method. This approach ensures distributed demand response in islanded microgrid networks, optimizing network efficiency and resilience. While still in its nascent stages, this technology does have potential to contribute to a more equitable distribution of energy in the pursuit of community development. Many developing countries still see regular

rolling blackouts due to an inability to produce adequate electricity, and this could be a powerful tool in ensuring intelligent energy distribution.

Part 3: Microgrid Policies and Subsidies

The previous sections discussed how markets impact microgrids, the feasibility of them, and how they can benefit communities. With any luck, my stance has been made clear that I believe this to be a technology worth promoting, and would now like to explore some of the ways this goal can be achieved. The paper titled "The Feasibility of Utilizing Second Life EV Batteries" (Mohammed, 2021) explores the potential of reusing degraded electric vehicle (EV) batteries, known as Second Life Batteries (SLB), focusing on their lifespan, economic benefits, and environmental impact. With the rapid growth in EVs, an estimated 85 million will be on the road by 2030, leading to a surplus of discarded batteries. These batteries, often at 70-80% capacity, can be repurposed, primarily in Energy Storage Systems (ESS), offering both environmental benefits and revenue opportunities. It reviewed various projects to understand SLB's current state, covering technical feasibility, economics, and environmental implications. SLBs are estimated to have an additional 7-10 years of lifespan in second-life applications, providing significant utility beyond their initial automotive use. Economically, while repurposing SLBs can be financially viable, challenges include the lack of standardized assessment and limited reliable data. Environmentally, utilizing SLBs could reduce CO₂ emissions by 56% compared to natural fuel gas, aligning with waste management strategies for a cleaner environment. These challenges can be greatly reduced via the implementation of supportive governmental assistance, both market based as well as policy based.

"Research on a Microgrid Subsidy Strategy Based on Operational Efficiency of the Industry Chain" (Long, 2018) also examines the role of government subsidies in promoting the development of microgrids. It highlights the necessity of subsidies in the early stages of microgrid development to balance costs and benefits. The central challenge addressed is optimizing the operational efficiency of the microgrid industry chain, considering various subsidy targets and methods. By analyzing this model, the study determines price and return indicators for each participant within the microgrid industry chain under different subsidy scenarios. The research contrasts and compares optimal subsidy strategies and influencing factors based on varying operational efficiency indicators. It also conducts a numerical analysis to validate and analyze the model, discussing the impacts of different development stages, technological levels, and subsidy amounts on the operational efficiency of the microgrid industry chain and the returns for each stakeholder involved.

Conclusion

It is my hope that one day soon, renewable energy based microgrids will become economically and practically feasible enough to see widespread adoption in both developed, and developing countries. While centralized energy has served us well throughout the years, there is undoubtedly great potential and benefit in also supporting the ability for communities to practice resilience via securing their energy generation methods.

The potential of microgrids to democratize energy, allowing communities to take charge of their

energy production and consumption, is an exciting prospect. This democratization can lead to enhanced community cohesion, as local stakeholders come together to manage a resource that is vital to their daily lives and economic well-being. In developing regions, where access to reliable electricity is still a challenge, microgrids could be a game-changer, providing the necessary impetus for economic development, improved healthcare, education, and overall quality of life.

However, the successful implementation and sustainability of microgrid projects hinge on thoughtful policies and supportive frameworks. Governments and policy makers play a crucial role in this. By offering subsidies, incentives, and regulatory support, they can lower the barriers to entry for microgrid deployment. Furthermore, integrating innovative ideas such as the use of second life EV batteries, as highlighted in Mohammed et al. (2021), could provide cost-effective solutions for energy storage in microgrids, enhancing their feasibility and appeal.

Citations

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